



Full Length Article

Clinical and laboratory diagnosis of rare coagulation disorders (RCDs)

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ARTICLE INFO

Keywords:

Rare coagulation disorders

Disease burden

Phenotype

Genotype

Diagnosis

ABSTRACT

Rare coagulation disorders (RCDs) are a group of diseases due to coagulation factors deficiency leading to life-long bleeding diathesis. The diagnosis of RCDs is challenging due to the limited knowledge of these disorders and the large heterogeneity of their bleeding patterns. The clinical symptoms of RCDs are extremely diverse in terms of bleeding type, site, severity, age at onset, and duration. The strength of the association between clotting factor activity level in plasma and clinical symptoms is also variable within each RCD.

The clinical evaluation of RCDs starts with a detailed collection of clinical history and has been facilitated by bleeding assessment tools, however their effectiveness in diagnosing RCDs requires further investigation. The following laboratory diagnosis of RCDs involves coagulation screening tests, including activated partial thromboplastin time, prothrombin time, and thrombin time. After ruling out the presence of an inhibitor by mixing studies, in case of abnormal results, the specific deficiency is identified by performing one-stage clotting assays using the specific factor-depleted plasmas as substrate. In fibrinogen and FXIII deficiencies coagulation screening tests are not informative, therefore additional tests are needed. Global assays have been developed and are thought to aid in patient management, however, they are not well standardized yet.

In addition to outlining the principles of clinical and laboratory diagnosis, this review explores molecular basis of RCDs and laboratory techniques for genetic analysis, and discusses the importance and effectiveness of quality control programs to ensure standardized laboratory results.

1. Introduction

Rare coagulation diseases (RCDs) cover deficiencies of coagulation factors such as fibrinogen, prothrombin, factor (F) V, FVII, FX, FXI, and FXIII, and the combined deficiency of FV + FVIII and vitamin K-dependent factors. However, this grouping is merely conventional because RCDs differ in terms of classification and clinical symptoms [1]. Prevalence of RCDs ranges from one in 500,000 in FVII deficiency to one in two to three million in FXIII deficiency [2,3] (Table 1).

RCDs are transmitted as autosomal recessive traits, therefore males and females are equally affected and their prevalence may be greater in regions with high rates of consanguineous marriage (such as Middle East, India, and Pakistan), which increases the probability that offspring will inherit both abnormal alleles carrying genetic variations that cause the disease [4–7]. Such countries are generally low-income [8], with economic constraints that may limit diagnosis and treatment of RCDs.

Because RCDs are rare and there is limited knowledge on their aetiology and pathogenesis, the description of the natural history can be difficult. The clinical picture of RCDs is characterized by bleeding

symptoms, which may range from absent or minimal to life-threatening events such as central nervous system (CNS) and musculoskeletal bleeding. In addition, bleeding patterns vary widely among each RCD and also within individuals with the same RCD.

Traditionally, RCDs were classified according to the hemophilia classification (severe: < 1% of normal activity, moderate: 1–5%, mild: 6–40%). However, it soon became apparent that a single classification for all RCDs may not be appropriate. In 2012, the European Network of Rare Bleeding Disorders (EN-RBD) explored the question using data collected from 592 patients [1]. The data showed that clinical bleeding symptoms were strongly associated with clotting factor levels only in fibrinogen, FII, FX, and FXIII deficiency and it was weakly correlated with FV and FVII activity levels. No association was found between bleeding symptoms and FXI activity. Despite a clear correlation between coagulation factor activity level and clinical bleeding severity, the combined FV + FVIII deficiency was mainly associated with mild or moderate clinical symptoms and rarely experience severe bleeding [1]. Due to a very limited number of patients with FII deficiency, a clear classification could not be defined. This implies that each RCD, unlike hemophilia A and B, has its own profile in terms of severity of clinical

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<https://doi.org/10.1016/j.thromres.2019.09.006>

Received 19 June 2019; Received in revised form 4 September 2019; Accepted 5 September 2019

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Table 1

Prevalence of coagulation protein deficiencies and the corresponding genes and chromosomes. Prevalence refers to the severe form of deficiencies due to homozygous or compound heterozygous mutations.

Deficiency	Prevalence	Gene and chromosome
Afibrinogenemia	1/1 million	<i>FGA, FGB, FGG</i> (all on 4q28)
Hypo- and dys-fibrinogenemia	Higher prevalence is expected, as heterozygosity is sufficient to cause the phenotype	
Hypo-, dys-prothrombinemia	1/2 million	<i>F2</i> (11p11–q12)
Factor V	1/1 million	<i>F5</i> (1q24.2)
Combined FV and FVIII	1/1 million	<i>LMAN1</i> (18q21.3–q22)
		<i>MCFD2</i> (2p21–p16.3)
Factor VII	1/500,000	<i>F7</i> (13q34)
Factor X	1/1 million	<i>F10</i> (13q34)
Factor XI	1/1 million	<i>F11</i> (4q35.2)
Factor XIII	1/3 million	<i>F13A1</i> (6p24–p25)
		<i>F13B</i> (1q31–q32.1)
Vitamin K-dependent factors	Reported in about 50 families	<i>GGCX</i> (2p12)
		<i>VKORC1</i> (16p11.2)

symptoms.

In the same year, the Scientific and Standardization Committee (SSC) of the International Society on Thrombosis and Haemostasis (ISTH), based on data from the literature, the results from the EN-RBD and three additional national and international registers, has proposed a new severity classification: severe deficiency (coagulant activity associated with spontaneous major bleeding); moderate deficiency (coagulant activity associated with mild spontaneous or triggered bleeding); mild deficiency (coagulant activity associated with a mostly asymptomatic disease course) [9]. This new clinical and laboratory severity classification system for each RCD may have some practical utility for most of them, with the exception of FXI deficiency. Because the strength of the association between clotting factor activity and clinical phenotype was highly variable in each RCD, the committee has proposed different thresholds for each disorder (Table 2).

The usefulness of this classification was recently highlighted in a description of different cases of FVII deficiency and a review of the relevant literature [10]. The authors urge health care professionals to use this severity system in order to appreciate the risk of patients with FVII deficiency, especially those with factor activity of < 10%, whose symptoms do not correlate with FVII clotting activity.

Despite the growing literature on RCDs, their rarity and heterogeneity often make it difficult to recognise and diagnose them, in particular for those physicians who do not routinely deal with this type of disease. The following review reports the principles of clinical and laboratory (phenotypic and genetic) diagnosis that requires a tight collaboration between physicians and laboratory specialists and discusses the importance and effectiveness of quality control programs to ensure standardized laboratory results.

Table 2

Severity classification system proposed by the Factor VIII and Factor IX of the ISTH [9].

Deficiency	Laboratory phenotype		
	Coagulant activity		
	Severe	Moderate	Mild
Fibrinogen	Undetectable clot	0.1–1 g/L	> 1 g/L
Prothrombin	Undetectable activity	≤ 10%	> 10%
Factor V	Undetectable activity	< 10%	≥ 10%
Combined FV and FVIII	< 20%	20–40%	> 40%
Factor VII	< 10%	10–20%	> 20%
Factor X	< 10%	10–40%	> 40%
Factor XIII	Undetectable activity	< 30%	≥ 30%

2. Clinical evaluation

Almost everyone either affected with bleeding disorders or not, experiences some form of bleeding during lifetime. To determine whether a person referred for a bleeding disorder is likely to have an RCD, the optimal assessment of the presence and severity of bleeding symptoms is of great importance and the collection of a detailed personal and family history is crucial.

2.1. Clinical manifestations

The most common bleeding symptom in RCDs, across all ages, is mucocutaneous bleeding [3,11] and excessive bleeding after surgery or dental procedures, after circumcision in the neonatal period, and during menstruation and pregnancy [12]. On the contrary, most life-threatening symptoms such as umbilical cord and CNS bleeding are prevalent with severe fibrinogen, FII, FX and FXIII deficiencies and have been reported in a few patients with severe FVII deficiency [13–16]. Gastrointestinal (GI) bleeding is common in severe FX deficiency, but less frequent in all other RCDs except some cases of FV + FVIII deficiency [15,17]. Soft tissue and joint bleeding have been reported with all RCDs but are relatively more common in afibrinogenemia and severe FII, FX and FXIII deficiencies [13–16]. In general, FXI deficiency is rarely associated with serious bleeding manifestations [18] and spontaneous bleeding is uncommon even at moderately reduced FXI activity levels (< 15–20%); at variance, bleeding may occur after trauma, surgery, or post-partum, especially in tissues with high fibrinolytic activity such as the nasopharynx and urogenital tract [18]. In combined FV + FVIII deficiency, symptoms are usually mild, the most frequent manifestations being easy bruising, epistaxis, and bleeding after dental extractions [17].

Significant bleeding manifestations usually show in homozygous or compound heterozygous patients, while heterozygous patients, with approximately half-normal levels of coagulation factors are usually asymptomatic. However, mucocutaneous and surgery-associated bleedings have been reported in 20% of heterozygotes, and post-traumatic haemarthrosis and hematomas are reported, albeit rarely, in FVII and FX deficiencies [12,15].

Other non-haemorrhagic clinical manifestations may be associated with certain RCDs. In afibrinogenemia and dysfibrinogenemia, there have been reports of arterial and venous thromboembolic complications, bone cysts, and splenic infarcts [13]. Myocardial infarction and venous thrombosis have occurred in patients with cardiovascular risk factors in FXI deficiency after FXI concentrate infusion [19–21]. In some cases of FX deficiency, various non-coagulation-related abnormalities, including thrombocytopenia-absent radius syndrome, mitral valve prolapse, hypertrophic cardiomyopathy, and hypercholesterolemia, has been reported [22], however, the concomitant presence of

these abnormalities may be coincidental occurrences rather than being associated to FX deficiency.

Particular attention should be placed on women with RCDs, who may experience gynaecologic bleeding in addition to the common bleeding symptoms [23]. Heavy menstrual bleeding is the most common symptom reported in > 50% of women with all deficiencies [24,25], and it may be associated to chronic iron deficiency anaemia with low levels of ferritin and haemoglobin or pain. Consequently, women's quality of life may be negatively impacted due to limitations in social activities, work, and education [26,27]. Beyond menorrhagia, affected females are at an increased risk of developing gynaecological comorbidities as haemorrhagic ovarian cysts, endometriosis, endometrial hyperplasia, polyps, and fibroids. Pregnancy and childbirth pose particular clinical challenges, and miscarriages, bleeding during pregnancy, and post-partum haemorrhage are common with certain deficiencies, as fibrinogen, FV and FXIII deficiencies [24,25,28].

Due to the large heterogeneity in bleeding phenotypes, the role of contributing factors other than the specific clotting factor deficiency needs to be explored, such as the fibrinolytic potential of the site of injury, additional source of functional FV from platelets, reduced levels of tissue factor pathway inhibitor associated with FV deficiency and the possible coexistence of other haemostatic defects. The latter might be studied nowadays with the access to the next-generation sequencing (NGS) analysis.

2.2. Bleeding assessment tools

Because some RCDs are often associated with mild bleeding, it can be a challenge to evaluate symptoms because the reporting and interpretation of bleedings are prone to subjectivity. Patients used to having bleeding episodes may overlook significant symptoms, while healthy subjects may give undue consideration to symptoms that are actually trivial. Patients who have experienced bleeding, in particular severe bleedings, since birth or infancy and whose family members also suffer from bleeding are likely to have an inherited bleeding disorder. In a few cases, bleeding symptoms may indicate the distinct bleeding disorder; umbilical cord, for example, usually points to afibrinogenemia or FXIII deficiency. When abnormal bleeding begins at a later age, or if there is a change in bleeding tendency, an acquired cause can often be identified.

Standardized diagnostic tools, that systematically assess bleeding symptoms thus aiding in the diagnosis of bleeding disorders, have attracted growing interest over the past decade [29–34]. Among these, Bleeding Assessment Tools (BATs) are questionnaires that methodically score each bleeding symptom on the basis of the patient's most severe episodes and combine them into a bleeding score (BS) to help distinguish between the absence and presence of a bleeding disorder. Different BATs assign different normal ranges for bleeding scores. These ranges should be carefully defined for target populations (e.g., paediatric), because the specificity of a BS depends on the chosen cut-off. For the ISTH-BAT [34], the proposed normal cut-offs are > 4 in adult males, > 6 in adult females, and > 3 in children [35].

In a few studies carried out on RCD patients using these existing tools, the results have been inconclusive [36–38]. It has to be underlined that many BATs were originally designed for more common bleeding disorders, such as von Willebrand disease (VWD), or to diagnose mild bleeding disorders, and may not provide optimal assessment for patients with RCDs. Palla et al. have designed a new BAT specifically tailored for RCDs, using data retrieved from the EN-RBD database to avoid patients' subjective interpretation of symptoms, especially in mild cases [39]. A bleeding score of 1.5 using this BAT was able to distinguish between patients with and without RCD, with a sensitivity of ~67% and specificity of ~74%; a negative correlation was observed between BS and factor activity level, especially in fibrinogen and FXIII deficiencies [39]. This BAT was designed to circumvent the well-known issues of age and sex by introducing them as covariates in the formula to calculate probability. However, its validity and reliability in RCDs

requires further evaluation.

From a clinical standpoint, a BAT provides a structured approach to the diagnosis of bleeding disorders [40–42]. Patients should be considered as possibly having a bleeding disorder if their BS is above the normality threshold. These patients should then undergo laboratory haemostatic evaluation. The use of a BAT, therefore, may standardize clinical practice and allow a uniform diagnostic approach. Because bleeding histories (and BS) are not sensitive enough to rule out the presence of a bleeding disorder in subjects with a negative history, there is debate as to the diagnostic value of BATs. On the other hand, they may be useful in predicting bleeding risk, particularly when an antihemorrhagic treatment is available. This is an area that requires more evidence-based data. While there are encouraging findings suggesting that a history of bleeding correlates with a higher risk of future bleeding in patients with VWD [43,44], the only available study that also includes patients diagnosed with RCDs does not confirm the usefulness of the ISTH-BAT score for predicting the risk of future bleeding [45]. However, that study involved a very heterogeneous population of individuals who visited a tertiary centre with a suspected bleeding disorder, therefore the possibility that the BAT may predict bleeding in well-defined patient groups could not be ruled out.

3. Laboratory diagnosis

3.1. Phenotype evaluation

The initial laboratory diagnosis of RCDs involves coagulation screening tests, including prothrombin time (PT) and activated partial thromboplastin time (aPTT). A prolonged PT value without prolongation of aPTT is suggestive of FVII deficiency. The reverse pattern (prolonged aPTT with normal PT) suggests FXI deficiency once FVIII, FIX, and FXII deficiencies have been ruled out. If both PT and aPTT are prolonged, additional analyses should be performed to test for deficiencies of combined FV and FVIII, FX, FV, prothrombin or fibrinogen (Table 3).

Prolongation of aPTT should be followed by mixing studies (50:50) to rule out the presence of an inhibitor. If the mixing study corrects the prolongation, the clotting factor deficiency should be identified by performing one-stage clotting assays using the specific factor-depleted plasmas as substrate, based on aPTT in the case of FXI and PT in the case of FII, FV, FVII, and FX [12].

In addition to PT and aPTT, thrombin clotting time (TT) should be measured to diagnose fibrinogen deficiency, if available. TT reflects the reaction between thrombin and fibrinogen and will therefore be prolonged in cases of very low fibrinogen levels (< 1.0 g/L), however, it is not required for the diagnosis of fibrinogen deficiency. On the contrary, the functional Clauss assay is the recommended test. In addition, factor antigenic assays are essential for diagnosing quantitative fibrinogen deficiency and FII deficiency, in order to appropriately classify and treat patients with dysfibrinogenemia and dysprothrombinaemia, both associated with an increased thrombotic risk [13,14,46].

Table 3

General features of autosomal recessive deficiency of coagulation factors.

Deficiency	APTT	PT	TT
Fibrinogen	↑↑	↑↑	↑↑
Afibrinogenemia	↑	↑	↑↑
Dys- and Hypodysfibrinogenemia	↑	↑	Normal
Prothrombin	↑	↑	Normal
Factor V	↑	↑	Normal
Combined factor V and VIII	↑	↑	Normal
Factor VII	Normal	↑	Normal
Factor X	↑	↑	Normal
Factor XI	↑	Normal	Normal
Factor XIII ^a	Normal	Normal	Normal
Vitamin-K dependent coagulation factors	↑	↑↑	Normal

^a Specific assays required.

Screening clotting tests (PT, aPTT, fibrinogen, platelet count, bleeding time) are normal in FXIII deficiency (Table 3), thus its diagnosis requires specific assays. FXIII deficiency has been most commonly diagnosed using clot solubility test including 5 M urea and monochloroacetic acid as solubilizing agents, but this is not a quantitative or standardized assay and only detects activity levels of < 5%, resulting in underdiagnosis of patients with reduced levels yet above 5% [47]. Therefore, this test is no longer recommended; FXIII activity should be assayed quantitatively measuring ammonia release during the transglutaminase reaction [48]. This test requires plasma blanking to avoid the FXIIIa-independent ammonia release that could lead to incorrect results in the low-activity range (< 5–10%) [49]. If FXIII activity is decreased, the subtype deficiency (FXIII-A or FXIII-B) needs to be determined with an immunological FXIII antigen assay to assure appropriate classification and treatment [50].

3.2. Global assays

As mentioned above, the most common standard tests for the diagnosis of clotting disorders are PT and PTT, as well as specific tests for fibrinogen and FXIII deficiency. These tests are fast and relatively easy to perform in any coagulation laboratory, and many laboratory manuals are available for guidance (e.g. [51–53]). However, because standard coagulation tests are limited to phases of the coagulation process, many researchers believe that the ideal coagulation test should evaluate the entire *in vivo* phenomenon.

Although such a test is not yet available, global assays are thought to aid in the individual phenotypic prediction and hence in the clinical management of the patient, in part because clotting level activity in patients with RCDs does not always correlate with the bleeding risk [1,54]. Several global assays have been developed in recent years and for two of them, thromboelastography (TEG) and thrombin generation assay (TGA), the literature is growing with regard to bleeding disorders [55–58] and RCDs [59–62], including for diagnostic aims [63,64]. Despite almost 20 years of research and several published papers on the use of global assays in bleeding disorders, there are still no standard protocols or consensus for clinical practice and diagnostics. To this aim, the Factor VIII, Factor IX, and Rare Coagulation Disorders of the ISTH has published recommendations for standardizing the methodology approaches for both TEG and TGA [65,66].

3.3. Molecular diagnosis

Molecular diagnosis is based on the identification of causative variants in genes encoding the corresponding coagulation factors located on various chromosomes (Table 1) [3]. Exceptions are combined FV + FVIII deficiency [67], caused by mutations in genes encoding the proteins involved in FV and FVIII intracellular transport (MCFD2 and LMAN1), and VKCFD [68], caused by variations in genes encoding the enzymes involved in post-translational modification and vitamin K metabolism (*GGCX* and *VKOR*). A variation on one of these genes may cause reduction of protein expression and/or production of a dysfunctional protein. The inheritance pattern is autosomal recessive for all RCDs, except for some cases of FXI and of hypo- and dysfibrinogenemia [69]. Information on the known gene variations causing RCDs is available through the ISTH mutation database (<http://www.isth.org/?MutationsRareBleedin>, accessed June 2019) and the Human Gene Mutation Database (<http://www.hgmd.cf.ac.uk/ac/index.php>, accessed June 2019).

Missense mutations are the most frequent cause of RCDs, accounting for 50–80% of known gene variations, except combined FV + FVIII deficiency due to mutations on *LMAN1* for which the most common gene variations (50%) are insertions/deletions [69]. Insertions/deletions represent 20–30% of gene variations of the fibrinogen genes (*FGA*, *FGB* and *FGG*), *F5*, *MCFD2*, and *F13A* genes and < 15% of the remaining coagulation factor genes. Splicing and nonsense mutations

comprise 5–15% of the identified mutations, with a maximum rate of 20% in *LMAN1*. Variants located in the 3' and 5' untranslated regions of genes are the least frequent (< 5%) and found only in fibrinogen, FVII, FXI, and FXIII deficiencies [69].

Sanger sequencing has been the most frequent technique of genotyping, but although it is widespread and cheaper than it was a few decades ago, it is still too slow and expensive. NGS is a new approach that enables the simultaneous investigation of multiple genes at a manageable cost. New initiatives, such as the ThromboGenomics project, aim to modernize the diagnosis of patients with rare inherited bleeding and platelet disorders through the development of a comprehensive genetic database based on the capacity of NGS. There are several advantages to using NGS with gene-specific panels and target capture sequencing as a diagnostic tool, including the simultaneous detection of single nucleotide variants, small duplications, and short insertions/deletions in many genes [70]. Recently, high-throughput sequencing platforms targeting a different number of genes relevant for bleeding disorders has been developed and tested in cases with and without previously known causal variants, with excellent sensitivity [71,72]. However, it is important to consider the limitations of this approach: mutations that are in deep introns, gross abnormalities, rearrangements, large deletions/insertions or total exon deletions that are not detected by sequencing-based technology may lead to a diagnosis failure [73]. In these cases, alternative approaches include multiplex ligation-dependent probe amplification (MLPA) for the identification of large deletions/duplications and long PCR for the detection of exon deletions [74].

Despite significant advances in knowledge, ~5% of patients with severe deficiencies have no identifiable genetic defect. For these individuals, the use of NGS and additional investigation of the deleterious/causative role of identified sequence variations may point to novel genetic pathways. For instance, exome resequencing, coupled with advanced bioinformatics algorithms, may help to identify new causal genes for autosomal recessive diseases and to arrive at an estimate of the likelihood that a variant will affect protein expression and/or function [75]. This method may only require a few unrelated patients to reach reliable results.

3.4. Quality control

The reliability of laboratory results depends on numerous factors, including knowledge of pre-analytical and analytical variables; adequate laboratory equipment, reagents and reference material; and protocols that should be as standardized as possible [51]. A suitable quality management system should be in place that monitors some of these components by means of internal quality control (IQC and external quality assurance (EQA) [76].

Laboratory scientists and clinicians should be aware of the choice of reagents and reference materials and performance of the material in use in their laboratory. PT and APTT reagents are available from numerous manufacturers and the reagents components may vary accordingly. For example, the source of tissue factor (TF) used in thromboplastin may vary between recombinant human or rabbit brain, depending upon manufacturer and brand. Similarly, the type of activator and phospholipid source in the APTT reagents vary according to manufacturer. The sensitivity of the screening test to clotting factors levels can vary accordingly between reagents brands and also between reagent lots of the same material.

Standardization of factor assays is assured by used of “certified reference material” for calibration of the assay. This should be metrologically traceable to an international reference material.

EQA providers, in addition to providing individual reports for participating laboratories, also provide reports on multicentre test performance and publication of specific areas of interest [77,78].

For example, the results of two different multicentre exercises were recently described in two publications circulated by the UK National

External Quality Assessment Scheme for Blood Coagulation (UK NEQAS BC) in close collaboration with the Prospective Rare Bleeding Disorders Database (PRO-RBDD) study group (<http://eu.rbdd.org/>). The first exercise concerned the approach taken by laboratories to investigate fibrinogen levels in samples from patients with dysfibrinogenemia and controls, and in one sample in which fibrinogen levels had been depleted to create an afibrinogenemic sample. Results showed that some widely used methodologies, using both optical and mechanical endpoint detection systems, may fail to detect the rare Longmont fibrinogen variant [79]. In the second exercise, samples from a FXIII-deficient subject and a normal donor were sent to participating centres to investigate for FXIII deficiency. According to the findings, about 70% of centres employ FXIII assays, complying with international recommendations. However, diagnostic errors were made by centres using both FXIII assays and solubility screen [80]. This conclusion reinforces the need for laboratories to ensure good quality assurance procedures to improve the accuracy of diagnoses.

4. Conclusions

Although RCDs working groups have proposed recommendations for measuring specific factor assays, the diagnosis of RCDs is still a challenge due to the lack of standardization of laboratory assays at a global level [81]. Participation in EQA programs is an important requirement in terms of standardization and assurance of the clinical suitability of laboratory measurements. Well-designed EQA programs can evaluate the reliability of measurement systems and the quality of results. The continuous participation in EQA programs, through comparison with other laboratories, sheds light on points of failure that could otherwise remain undetected and lead to incorrect or late diagnosis. This is especially important in countries where RCDs are more frequent [82]. The promotion of collaborative projects and twinning programs between expert and less expert centres could be successful in increasing the diagnosis of RCDs and therefore in improving patient management.

Author contributions

Both authors conceived and wrote the manuscript.

Declaration of competing interest

MM reports travel support from Pfizer; RP reports travel support from Grifols and Pfizer and speaker fees from Novonordisk.

References

- [1] F. Peyvandi, R. Palla, M. Menegatti, et al., European Network of Rare Bleeding Disorders Group. Coagulation factor activity and clinical bleeding severity in rare bleeding disorders: results from the European Network of Rare Bleeding Disorders, *J Thromb Haemost.* 10 (2012) 615–621.
- [2] P.M. Mannucci, S. Duga, F. Peyvandi, Recessively inherited coagulation disorders, *Blood.* 104 (2004) 1243–1252.
- [3] F. Peyvandi, F. R. Palla, M. Menegatti, P.M. Mannucci, Introduction. Rare bleeding disorders, General aspects of clinical features, diagnosis, and management, *Semin Thromb Hemost.* 35 (2009) 349–355.
- [4] M. Borhany, M. Z. Pahore, Z. Ul Qadr, et al., Bleeding disorders in the tribe: result of consanguineous in breeding, *Orphanet J Rare Dis.* 5 (2010) 23.
- [5] I.C. Jaouad, S.C. Elalaoui, A. Sbati, et al., Consanguineous marriages in Morocco and the consequence for the incidence of autosomal recessive disorders, *J Biosoc Sci.* 41 (2009) 575–581.
- [6] M. Saadat, M. Ansari-Lari, D.D. Farhud, Consanguineous marriage in Iran, *Ann Hum Biol.* 31 (2004) 263–269.
- [7] H. Peretz, A. Mulai, S. Usher, et al., The two common mutations causing factor XI deficiency in Jews stem from distinct founders: one of ancient Middle Eastern origin and another of more recent European origin, *Blood.* 90 (1997) 2654–2659.
- [8] Regional and income groupings. WHO regional groupings https://www.who.int/gho/publications/world_health_statistics/EN_WHS2015_Annex.pdf?ua=1, (accessed 9 May 2019).
- [9] F. Peyvandi, D. Di Michele, P.H. Bolton-Maggs, C.A. Lee, A. Tripodi, A. Srivastava; Project on Consensus Definitions in Rare Bleeding Disorders of the Factor VIII/Factor IX Scientific and Standardisation Committee of the International Society on Thrombosis and Haemostasis, Classification of rare bleeding disorders (RBDs) based on the association between coagulant factor activity and clinical bleeding severity, *J Thromb Haemost.* 10 (2012) 1938–1943.
- [10] S. Jain, J. Donkin, M.J. Frey, S. Peltier, S. Gunawardena, D.L. Cooper, Phenotypic variability in congenital FVII deficiency follows the ISTH-SSC severity classification guidelines: a review with illustrative examples from the clinic, *Journal of Blood Medicine.* 9 (2018) 211–218.
- [11] S.S. Acharya, A. Coughlin, D.M. Dimichele, North American Rare Bleeding Disorder Study G. Rare Bleeding Disorder Registry: deficiencies of factors II, V, VII, X, XIII, fibrinogen and dysfibrinogenemias, *J Thromb Haemost.* 2 (2004) 248–256.
- [12] R. Palla, F. Peyvandi, A.D. Shapiro, Rare bleeding disorders: diagnosis and treatment, *Blood.* 125 (2015) 2052–2061.
- [13] P. de Moerloose, A. Casini, M. Neerman-Arbez, Congenital fibrinogen disorders: an update, *Semin Thromb Hemost.* 39 (2013) 585–595.
- [14] S. Lancellotti, M. Basso, R. De Cristofaro, Congenital prothrombin deficiency: an update, *Semin Thromb Hemost.* 39 (2013) 596–606.
- [15] M. Menegatti, F. Peyvandi, Factor X deficiency, *Semin Thromb Hemost.* 35 (2009) 407–415.
- [16] V. Schroeder, H.P. Kohler, Factor XIII Deficiency: An Update, *Semin Thromb Hemost.* 39 (2013) 632–641.
- [17] C. Zheng, B. Zhang, Combined deficiency of coagulation factors V and VIII: an update, *Semin Thromb Hemost.* 39 (2013) 613–620.
- [18] S. Duga, S. O. Salomon, Congenital Factor XI Deficiency: An Update, *Semin Thromb Hemost.* 39 (2013) 621–631.
- [19] P.H. Bolton-Maggs, Factor XI deficiency and its management, *Haemophilia.* 6 (Suppl. 1) (2000) 100–109.
- [20] J.C. Meijers, W.L. Tekelenburg, B.N. Bouma, R.M. Bertina, F.R. Rosendaal, High levels of coagulation factor XI as a risk factor for venous thrombosis, *N Engl J Med.* 342 (2000) 696–701.
- [21] O. Salomon, D.M. Steinberg, R. Dardik, et al., Inherited factor XI deficiency confers no protection against acute myocardial infarction, *J Thromb Haemost.* 1 (2003) 658–661.
- [22] A. Girolami, E. Ruzzon, F. Tezza, R. Scandellari, M. Scapin, P. Scarparo, Congenital FX deficiency combined with other clotting defects or with other abnormalities: a critical evaluation of the literature, *Haemophilia.* 14 (2008) 323–328.
- [23] A.H. James, More than menorrhagia: a review of the obstetric and gynaecological manifestations of bleeding disorders, *Haemophilia.* 11 (2005) 295–307.
- [24] A.H. James, Women and bleeding disorders, *Haemophilia.* 16 Suppl 5 (2010) 160–167.
- [25] C.A. Lee, C. Chi, S.R. Pallord, et al., The obstetric and gynaecological management of women with inherited bleeding disorders—review with guidelines produced by a taskforce of UK Haemophilia Centre Doctors' Organization, *Haemophilia.* 12 (2006) 301–336.
- [26] R. Kulkarni, Improving care and treatment options for women and girls with bleeding disorders, *Eur J Haematol.* 95 Suppl 81 (2015) 2–10.
- [27] P.A. Kouides, P.D. Phatak, P. Burkart, et al., Gynaecological and obstetrical morbidity in women with type I von Willebrand disease: results of a patient survey, *Haemophilia.* 6 (2000) 643–648.
- [28] A. Inbal, L. Muszbek, Coagulation factor deficiencies and pregnancy loss, *Semin Thromb Hemost.* 29 (2003) 171–174.
- [29] J.M. Higham, P.M. O'Brien, R.W. Shaw, Assessment of menstrual blood loss using a pictorial chart, *Br J Obstet Gynaec.* 97 (1990) 734–739.
- [30] A. Srámek, J.C. Eikenboom, E. Briët, J.P. Vandenbroucke, F.R. Rosendaal, Usefulness of patient interview in bleeding disorders, *Arch Inter Med.* 155 (1995) 1409–1415.
- [31] J.E. Sadler, F. Rodeghiero, Provisional criteria for the diagnosis of VWD type 1, *J Thromb Haemost.* 3 (2005) 775–777.
- [32] F. Rodeghiero, G. Castaman, A. Tosetto, et al., The discriminant power of bleeding history for the diagnosis of type 1 von Willebrand disease: an international, multicenter study, *J Thromb Haemost.* 3 (2005) 2619–2626.
- [33] M. Bowman, G. Mundell, J. Grabell, et al., Generation and validation of the Condensed MCMDM-1VWD Bleeding Questionnaire for von Willebrand disease, *J Thromb Haemost.* 6 (2008) 2062–2066.
- [34] F. Rodeghiero, A. Tosetto, T. Abshire, et al., ISTH/SSC bleeding assessment tool: a standardized questionnaire and a proposal for a new bleeding score for inherited bleeding disorders, *J Thromb Haemost.* 8 (2010) 2063–2065.
- [35] M. Elbatarny, S. Mollah, J. Grabell, et al., Zimmerman Program Investigators, Normal range of bleeding scores for the ISTH-BAT: adult and pediatric data from the merging project, *Haemophilia.* 20 (2014) 831–835.
- [36] S.M. Siboni, M. Spreafico, L. Calò, et al., Gynaecological and obstetrical problems in women with different bleeding disorders, *Haemophilia.* 15 (2009) 1291–1299.
- [37] A. Kulkarni, C.A. Lee, A. Griffeon, R.A. Kadir, Disorders of menstruation and their effect on the quality of life in women with congenital factor VII deficiency, *Haemophilia.* 12 (2006) 248–252.
- [38] S.E. Shapiro, E. Phillips, R.A. Manning, et al., Clinical phenotype, laboratory features and genotype of 35 patients with heritable dysfibrinogenemia, *Br J Haematol.* 160 (2013) 220–227.
- [39] R. Palla, S.M. Siboni, M. Menegatti, K.M. Musallam, F. Peyvandi, European Network of Rare Bleeding Disorders, Establishment of a bleeding score as a diagnostic tool for patients with rare bleeding disorders, *Thromb Res.* 148 (2016) 128–134.
- [40] F.C.J.I. Moenen, P.J. Nelemans, S.E.M. Schols, H.C. Schouten, Y.M.C. Henskens, E.A.M. Beckers, The diagnostic accuracy of bleeding assessment tools for the identification of patients with mild bleeding disorders: a systematic review, *Haemophilia.* 24 (2018) 525–535.
- [41] A. Tosetto, Bleeding assessment tools: limits and advantages for the diagnosis and

- prognosis of inherited bleeding disorders, *Sem Thromb Haemost.* 42 (2016) 463–470.
- [42] J. Boender, M.J. Kruij, F.W. Leebeek, A diagnostic approach to mild bleeding disorders, *J Thromb Haemost.* 14 (2016) 1507–1516.
- [43] G. Castaman, A.B. Federici, A. Tosetto A, et al., Different bleeding risk in type 2A and 2M von Willebrand disease: a 2-year prospective study in 107 patients, *J Thromb Haemost.* 10 (2012) 632–638.
- [44] A.B. Federici, P. Bucciarelli, G. Castaman, et al., The bleeding score predicts clinical outcomes and replacement therapy in adults with von Willebrand disease, *Blood.* 123 (2014) 4037–4044.
- [45] M.R. Fasulo, E. Biguzzi, M. Abbattista, et al., The ISTH Bleeding Assessment Tool and the risk of future bleeding, *J Thromb Haemost.* 16 (2018) 125–130.
- [46] A. Casini, A. Undas, R. Palla, J. Thachil, P. de Moerloose, for the Subcommittee on Factor XIII and Fibrinogen, Diagnosis and classification of congenital fibrinogen disorders: communication from the SSC of the ISTH, *J Thromb Haemost.* 16 (2018) 1887–1890.
- [47] A. Dorgalaleh, S. Tabibian, V. Assadollahi, et al., Comparison of 2 Methods of Clot Solubility Testing in Detection of Factor XIII Deficiency, *Lab Med.* 47 (2016) 283–285.
- [48] L. Muszbek, Z. Bagoly, A. Cairo, F. Peyvandi, Novel aspects of factor XIII deficiency, *Curr Opin Hematol.* 18 (2011) 366–372.
- [49] E. Ajzner, L. Muszbek, Kinetic spectrophotometric factor XIII activity assays: the subtraction of plasma blank is not omissible, *J Thromb Haemost.* 2 (2004) 2075–2077.
- [50] H.P. Kohler, A. Ichinose, R. Seitz, et al., Factor XIII and Fibrinogen SSC Subcommittee of the ISTH, Diagnosis and classification of factor XIII deficiencies, *J Thromb Haemost.* 9 (2011) 1404–1406.
- [51] S. Kitchen, A. McCraw, M. Echenagucia, *Diagnosis of Hemophilia and Other Bleeding Disorders. A Laboratory Manual*, 2nd ed. <https://www.wfh.org/en/page.aspx?pid=877> (accessed 9 May 2019).
- [52] A. Dorgalaleh, S. Tabibian, S. Hosseini, M. Shamsizadeh, Guidelines for laboratory diagnosis of factor XIII deficiency, *Blood Coagul Fibrinolysis.* 27 (2016) 361–364.
- [53] I.J. Mackie, S. Kitchen, S.J. Machin, G.D. Lowe, Haemostasis and Thrombosis Task Force of the British Committee for Standards in Haematology, Guidelines on fibrinogen assays, *Br J Haematol.* 121 (2003) 396–404.
- [54] H.M. van den Berg, P.H.G. De Groot, K. Fischer, Phenotypic heterogeneity in severe hemophilia, *J Thromb Haemost.* 5 Suppl 1 (2007) 151–156.
- [55] M. Chitlur, G. Young, Global assays in hemophilia, *Semin Hematol.* 53 (2016) 40–45.
- [56] M. Lak, B. Scharling, A. Blemings, et al., Evaluation of rFVIIa (NovoSeven) in Glanzmann patients with thromboelastogram, *Haemophilia.* 14 (2008) 103–110.
- [57] Y. Dargaud, A. Lienhart, S. Meunier, et al., Major surgery in a severe haemophilia A patient with high titre inhibitor: use of the thrombin generation test in the therapeutic decision, *Haemophilia.* 11 (2005) 552–558.
- [58] K. Varadi, C. Negrier, E. Berntorp, et al., Monitoring the bioavailability of FEIBA with a thrombin generation assay, *J Thromb Haemost.* 1 (2003) 2374–2380.
- [59] A.E. Lovejoy, T.C. Reynolds, J.E. Visich, et al., Safety and pharmacokinetics of recombinant factor XIII-A2 administration in patients with congenital factor XIII deficiency, *Blood.* 108 (2006) 57–62.
- [60] C. Bulato, C. Novembrino, M.B. Anzoletti, et al., "In vitro" correction of the severe factor V deficiency-related coagulopathy by a novel plasma-derived factor V concentrate, *Haemophilia.* 24 (2018) 648–656.
- [61] G.N. Pike, A.M. Cumming, J. Thachil, et al., Evaluation of the use of global haemostasis assays to monitor treatment in factor XI deficiency, *Haemophilia.* 23 (2017) 273–283.
- [62] M. Van Geffen, M. Menegatti, A. Loof, et al., Retrospective evaluation of bleeding tendency and simultaneous thrombin and plasmin generation in patients with rare bleeding disorders, *Haemophilia.* 18 (2012) 630–638.
- [63] R. Al Dieri, F. Peyvandi, E. Santagostino, et al., The thrombogram in rare inherited coagulation disorders: its relation to clinical bleeding, *Thromb Haemost.* 88 (2002) 576–582.
- [64] L.A. Parunov, S.S. Surov, Y. Liang, et al., Can the diagnostic reliability of the thrombin generation test as a global haemostasis assay be improved? The impact of calcium chloride concentration, *Haemophilia.* 23 (2017) 466–475.
- [65] Y. Dargaud, A.S. Wolberg, E. Gray, et al., Factor VIII, Factor IX, and Rare Coagulation Disorders Subcommittee of the Scientific and Standardisation Committee of the International Society on Thrombosis and Haemostasis, Proposal for standardized preanalytical and analytical conditions for measuring thrombin generation in hemophilia: communication from the SSC of the ISTH, *J Thromb Haemost.* 15 (2017) 1704–1707.
- [66] M. Chitlur, G.E. Rivard, D. Lillicrap, et al., Factor VIII, Factor IX, and Rare Coagulation Disorders Subcommittee of the Scientific and Standardisation Committee of the International Society on Thrombosis and Haemostasis, Recommendations for performing thromboelastography/thromboelastometry in hemophilia: communication from the SSC of the ISTH, *J Thromb Haemost.* 12 (2014) 103–106.
- [67] B. Zhang, M.A. Cunningham, W.C. Nichols, et al., Bleeding due to disruption of a cargo-specific ER-to-Golgi transport complex, *Nat Genet.* 34 (2003) 220–225.
- [68] B. Brenner, A.A. Kuperman, M. Watzka, et al., Vitamin K-dependent coagulation factors deficiency, *Semin Thromb Hemost.* 35 (2009) 439–446.
- [69] F. Peyvandi, T. Kunicki, D. Lillicrap, Genetic sequence analysis of inherited bleeding diseases, *Blood.* 122 (2013) 3423–3431.
- [70] B. Sikkema-Raddatz, L.F. Johansson, E.N. Boer, et al., Targeted next-generation sequencing can replace Sanger sequencing in clinical diagnostics, *Hum Mutat.* 34 (2013) 1035–1042.
- [71] I. Simeoni, J.C. Stephens, F. Hu, et al., A high-throughput sequencing test for diagnosing inherited bleeding, thrombotic, and platelet disorders, *Blood.* 127 (2016) 2791–2803.
- [72] J.M. Bastida, M. Del Rey, M.L. Lozano, et al., Design and application of a 23-gene panel by next-generation sequencing for inherited coagulation bleeding disorders, *Haemophilia.* 22 (2016) 590–597.
- [73] J.M. Weaver, P.A. Edwards, Targeted next-generation sequencing for routine clinical screening of mutations, *Genome Med.* 3 (2011) 58.
- [74] R. Daber, S. Sukhadia, J.J. Morrisette, Understanding the limitations of next-generation sequencing informatics, an approach to clinical pipeline validation using artificial data sets, *Cancer Genet.* 206 (2013) 441–448.
- [75] F. Peyvandi, F. C.P. Hayward, Genomic approaches to bleeding disorders, *Haemophilia.* 22 Suppl 5 (2016) 42–45.
- [76] F. Braga, S. Pasqualetti, M. Panteghini, The role of external quality assessment in the verification of in vitro medical diagnostics in the traceability era, *Clin Biochem.* 57 (2018) 23–28.
- [77] E.J. Favaloro, I. Jennings, J. Olson, et al., Towards harmonization of external quality assessment/proficiency testing in hemostasis, *Clin Chem Lab Med.* 57 (2018) 115–126.
- [78] N.D. Zantek, C.P. Hayward, T.G. Simcox, et al., An assessment of the state of current practice in coagulation laboratories, *Am J Clin Pathol.* 146 (2016) 378–383.
- [79] I. Jennings, S. Kitchen, M. Menegatti, et al., Potential misdiagnosis of dysfibrinogenemia: Data from multicentre studies amongst UK NEQAS and PRO-RBDD project laboratories, *Int J Lab Hematol.* 39 (2017) 653–662.
- [80] I. Jennings, S. Kitchen, M. Menegatti, et al., Detection of Factor XIII deficiency: data from multicentre exercises amongst UK NEQAS and PRO-RBDD project laboratories, *Int J Lab Hematol.* 39 (2017) 350–358.
- [81] S. Jain, S.S. Acharya, Management of rare coagulation disorders in 2018, *Transfus Apher Sci.* 57 (2018) 705–712.
- [82] M. Karimi, S. Haghpanah, A. Amirhakimi, et al., Spectrum of inherited bleeding disorders in southern Iran, before and after the establishment of comprehensive coagulation laboratory, *Blood Coagul Fibrinolysis.* 20 (2009) 642–645.